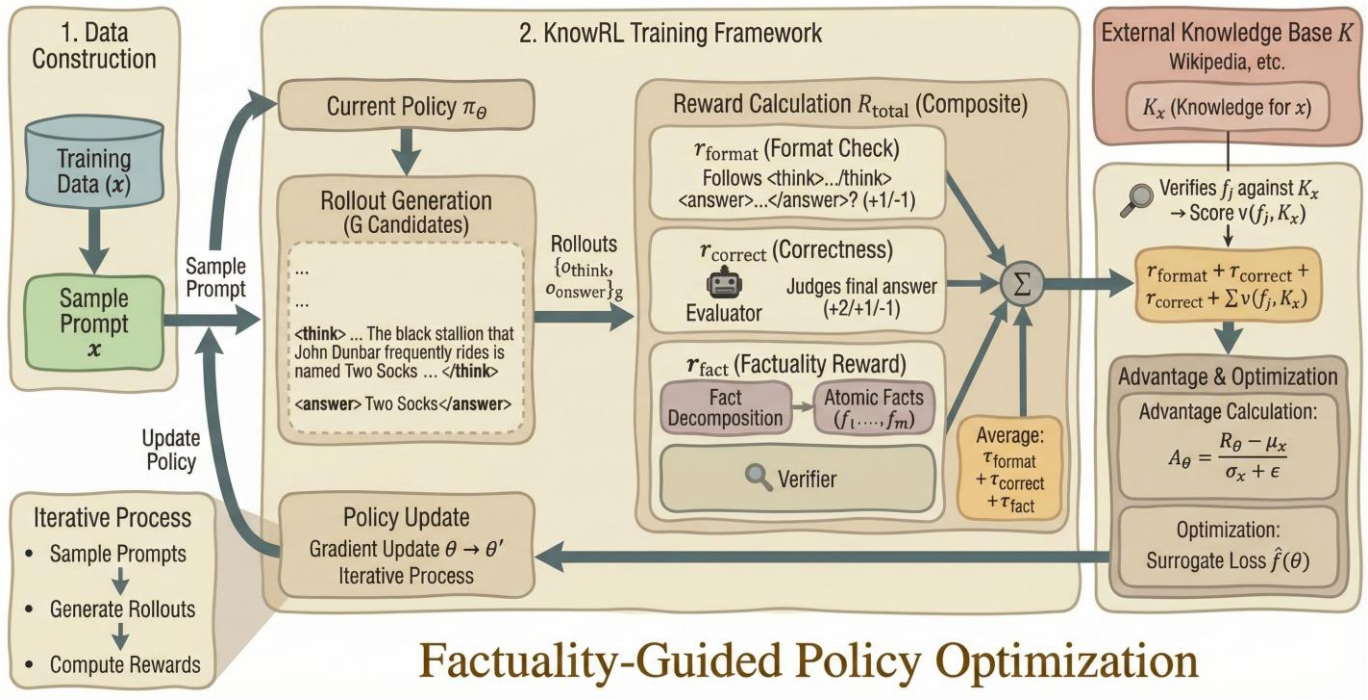

1. Data Construction
Training Data (x)
Sample Prompt x
Sample Prompt
2. KnowRL Training Framework
Current Policy π_θ
Rollout Generation (G Candidates)
...
...
<think> ... The black stallion that John Dunbar frequently rides is named Two Socks ... </think>
<answer> Two Socks</answer>
Rollouts $\{o_{\text{think}}, o_{\text{answer}}\}_g$
Reward Calculation R_{total} (Composite)
r_{format} (Format Check)
Follows <think>.../think> <answer>...</answer>? (+1/-1)
r_{correct} (Correctness)
Evaluator
Judges final answer (+2/+1/-1)
r_{fact} (Factuality Reward)
Fact Decomposition
Atomic Facts $(f_1, \ldots, f_m)$
Verifier
Average: $\tau_{\text{format}} + \tau_{\text{correct}} + \tau_{\text{fact}}$
External Knowledge Base K
Wikipedia, etc.
K_x (Knowledge for x)
Verifies f_j against K_x → Score $v(f_j, K_x)$
$r_{\text{format}} + \tau_{\text{correct}} + r_{\text{correct}} + \sum v(f_j, K_x)$
Advantage & Optimization
Advantage Calculation:
$A_\theta = \frac{R_\theta - \mu_x}{\sigma_x + \epsilon}$
Optimization:
Surrogate Loss $\hat{f}(\theta)$
Update Policy
Policy Update
Gradient Update $\theta \rightarrow \theta'$
Iterative Process
Iterative Process
• Sample Prompts
• Generate Rollouts
• Compute Rewards
Factuality-Guided Policy Optimization